

2023

JOURNEYMAN

OPEN

BOOK

FINAL EXAM #1



70 QUESTIONS

TIME LIMIT - 3 HOURS

TIME SPENT **MINUTES**

SCORE **%**

Journeyman Final Exam #1

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70 Questions - Time Limit 3 Hours

1. 60% is equivalent to ____.

- (a) 5/8 (b) 3/5 (c) 2/5 (d) 5/16

2. Each section of equipment that requires rear or side access to make field connections shall be so marked by the manufacturer on the ____.

- (a) front (b) right side (c) left side (d) rear

3. In other than one and two family dwellings, a receptacle outlet for indoor service equipment shall be located within ____ of the service equipment.

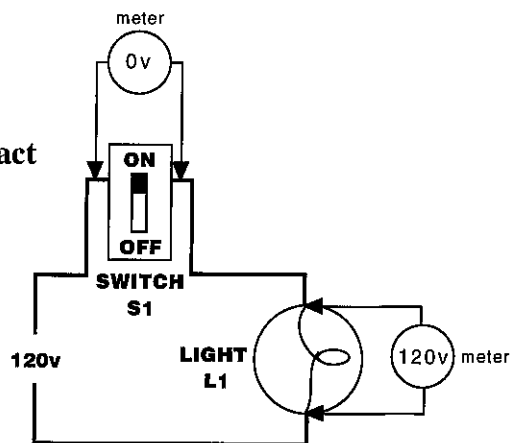
- (a) 25 feet (b) 50 feet (c) 75 feet (d) the same room or area

4. Disregarding demand factors, the calculated lighting load for a 5,000 sq.ft. office building is ____ volt-amperes.

- (a) 16,500 (b) 15,400 (c) 8,000 (d) 6,500

5. Refer to the figure below. GIVEN: Switch S1 is in the "ON" position, but the light does not come on. Voltage across L1 is measured to be 120 volts. Voltage across S1 is measured to be 0 volts. The light does not come on because ____.

- (a) the light is open (bulb burned out)
(b) the light and switch are shorted
(c) the light is good but the switch does not make contact
(d) there is a break in the wiring of the circuit



6. Decorative lighting and similar accessories used for holiday lighting and similar purposes shall be listed and ____.

- (a) marked (b) approved (c) labeled (d) stamped

7. Insulated conductors used inside switchgear or switchboards are required to be ____.

- (a) suitable (b) labeled (c) acceptable (d) listed

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8. Balconies, decks and porches that are within 4' horizontally of the dwelling unit shall have at least one receptacle outlet accessible from the balcony, deck, or porch. The receptacle outlet shall not be located more than ____ above the balcony, deck, or porch walking surface.

- (a) 36" (b) 48" (c) 60" (d) 78"

9. An acceptable color for ungrounded conductors is ____.

- (a) green (b) gray (c) pink (d) white

10. The calculated load of a 12 foot length of fixed multioutlet assembly in a commercial facility is ____ volt-amperes if the appliances it supplies are not likely to be used at the same time.

- (a) 1000 (b) 720 (c) 540 (d) 380

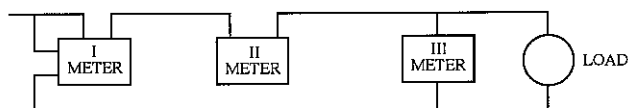
11. In a dwelling bedroom, any wall space ____ or more in width (including space measured around corners) and unbroken along the floor line by doorways and similar openings, fireplaces, and fixed cabinets that do not have countertops or similar work surfaces.

- (a) 18" (b) 24" (c) 30" (d) 36"

12. A wall-mounted central vacuum assembly connected to a single receptacle located in an attached garage shall be provided with ____ protection for personnel.

- (a) LCDI (b) GFCI (c) AFCI (d) both AFCI and GFCI

13. Which of the following is an ammeter?



- (a) I only (b) II only (c) III only (d) I, II or III

14. What is the ampacity of a #12 THWN conductor installed in an ambient temperature of 86°F if the raceway contains four current-carrying conductors?

- (a) 17.5a (b) 20a (c) 23a (d) 26a

15. If festoon lighting is installed without a messenger, the smallest allowable overhead conductor is ____ AWG.

- (a) #10 (b) #12 (c) #14 (d) #16

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16. For temporary holiday lighting, trees shall be permitted for supporting overhead spans of conductors and cables if the overhead wiring is arranged with _____, tension take-up devices, or other approved means to avoid damage from live vegetation.

(a) fittings (b) cable ties (c) strain relief devices (d) overhead clamps

17. The Code requires branch circuits be rated in accordance with the overcurrent protective device. Therefore, which of the following combinations are permitted for lighting branch circuits?

(a) 20, 25, 35, 40 and 50 amperes (b) 20, 25, 30, 40 and 50 amperes
(c) 15, 25, 30, 40 and 50 amperes (d) 10, 15, 20, 30, 40 and 50 amperes

18. Duty on elevator and dumbwaiter driving machine motors and driving motors of motor-generators used with generator field control shall be rated as _____.

(a) intermittent (b) lockable (c) continuous (d) varying

19. What does the alpha character I represent when stating the question $W = E \times I$?

(a) Intrinsic circuit (b) Intrinsic electromotive force
(c) Intensity of current (d) Isotopic character

20. Where multiple driven ground rods are used to form the grounding electrode system, in order to maintain an effective grounding electrode system, they shall not be less than _____ apart.

(a) 36" (b) 48" (c) 60" (d) 72"

21. The demand load for a 14 kW range installed in a dwelling unit is _____ kW.

(a) 8 (b) 8.4 (c) 8.8 (d) 12

22. Where a dwelling has a two car attached garage, the Code requires a minimum of _____ receptacle outlet(s) shall be installed in the garage.

(a) none required (b) one (c) two (d) three

23. Where ungrounded conductors are run in parallel in multiple raceways, the equipment grounding conductor, where used, shall be _____.

(a) installed in one raceway only (b) run in parallel in each raceway
(c) not required (d) bare

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24. All 15 or 20 amp, single-phase, 125 volt through 250 volt receptacles located within ____ feet of a fountain edge shall be provided with GFCI protection.

- (a) 20 (b) 24 (c) 25 (d) 30

25. For an arc welder with an I_{max} rating of 43 amps, what is the maximum rating for the overcurrent device supplying the welder?

- (a) 85 amp (b) 90 amp (c) 95 amp (d) 100 amp

26. The conductor is marked RHW-2 on the insulation, what does the -2 represent?

- (a) The cable has two conductors.
(b) The conductor is double insulated.
(c) The conductor is thermoplastic.
(d) The conductor has a maximum operating temperature of 90°C.

27. The ampacity of three #10 THWN-2 conductors installed in a raceway is ____ amps if the ambient temperature is 112°F.

- (a) 31.6 (b) 34.8 (c) 35 (d) 37.2

28. Fuel dispensing systems shall be provided with one or more clearly identified emergency shutoff devices or electrical disconnects. Such disconnects or devices shall be installed in approved locations but not less than 20 feet or more than ____ feet from the fuel dispensing devices that they serve.

- (a) 50 (b) 75 (c) 80 (d) 100

29. Where conduits enter a floor-standing switchboard, switchgear or, panelboard at the bottom, the conduits, including their end fittings, shall not rise more than ____ inches above the bottom of the enclosure.

- (a) 2 (b) 3 (c) 4 (d) 6

30. Thermostatically controlled switching devices serving both controllers and disconnecting means for fixed electric space heating equipment shall ____.

- (a) not be permitted
(b) be located not more than 8 feet above floor level
(c) open all grounded conductors when placed in the off position
(d) be designed so that the circuit cannot be energized automatically after the device has been manually placed in the off position

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31. In dwelling units, hallways of ____ feet or more in length shall have at least one receptacle outlet.

- (a) 6 (b) 8 (c) 10 (d) 12**

32. For a feeder tap not exceeding 10 feet in length and field installation, the maximum overcurrent device rating supplying a tap conductor with an ampacity of 40 amps is ____.

- (a) 150a (b) 200a (c) 350a (d) 400a**

33. A three-way switch is equivalent to a ____ switch.

- (a) DPST (b) DPDT (c) SPST (d) SPDT**

34. Which of the following cord types is permitted in wet location and is sunlight resistant?

- (a) SPT-2 (b) STOOW (c) THWN (d) XHWN**

35. One or more metal in-ground support structure(s) in direct contact with the earth vertically for ____ feet or more, with or without concrete encasement is a permitted grounding electrode.

- (a) 6 (b) 8 (c) 10 (d) 12**

Half way through the exam

36. For a service rated 100 through 400 amps, the service conductors supplying the entire load of a one family dwelling shall be permitted to have an ampacity of ____ of the service rating.

- (a) 83% (b) 80% (c) 75% (d) 70%**

37. Nonmetallic cable trays shall be made of ____ material.

- (a) watertight (b) waterproof (c) fire-resistant (d) flame-retardant**

38. Type UF cable is permitted to be used ____.

- (a) as service entrance (b) in commercial garages**
(c) where subject to physical damage (d) for wiring in wet, dry, or corrosive locations

39. A controller that includes motor overload protection ____ for group motor application shall be marked with the motor overload protection and the maximum branch-circuit short-circuit and ground-fault protection for such applications.

- (a) required (b) recognized (c) permitted (d) suitable**

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40. The permitted demand factor for five household clothes dryers in a multifamily dwelling is ____.

- (a) 70% (b) 75% (c) 80% (d) 85%

41. A stop switch is wired ____ in a motor circuit.

- (a) series (b) series-shunt (c) series-parallel (d) parallel

42. All 15 and 20 amp, 125 and 250 volt, nonlocking receptacles located in a wet location shall be listed ____ type.

- (a) weather proof (b) water proof (c) water resistant (d) weather resistant

43. Where no GFCI protection is provided, the mounting height of a paddle fan located above a spa or hot tub shall not be less than ____ feet.

- (a) 6 (b) 8 (c) 10 (d) 12

44. An unintentional, electrically conducting connection between an ungrounded conductor of an electrical circuit and the normally non-current-carrying conductors, metallic enclosures, metallic raceways, metallic equipment or earth is referred to as a ____.

- (a) ground fault (b) open circuit (c) short circuit (d) circuit bypass

45. Residential in sink electrically operated kitchen waste disposers shall be permitted to be cord-and-plug connected however, the flexible cord is to be not less than 18" in length and not over ____ in length.

- (a) 24" (b) 30" (c) 36" (d) 48"

46. 40% is equivalent to ____.

- (a) 5/8 (b) 3/5 (c) 2/5 (d) 5/16

47. Which of the following is the symbol for a temperature actuated switch?



(a)



(b)



(c)



(d)

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48. A 125 volt, 15 amp rated receptacle located in a hallway of a dwelling unit is required to be ____.

- (a) GFCI protected (b) on a dedicated circuit
(c) need a 20 amp receptacle (d) listed tamper-resistant

49. Direct-buried cables located in a trench below 2" of concrete shall have a minimum cover of ____.

- (a) 6" (b) 12" (c) 18" (d) 24"

50. Communications wires and cables and CATV type coaxial cables shall have a vertical clearance of not less than ____ from all points of roofs which they pass.

- (a) 18" (b) 36" (c) 6 feet (d) 8 feet

51. Personnel doors where equipment rated 800 amperes or more that contains overcurrent devices, is installed and there is a personnel door(s) intended for entrance to and egress from the working space less than ____ feet from the nearest edge of the working space.

- (a) 12 (b) 15 (c) 20 (d) 25

52. Outdoor mobile home disconnecting means shall be installed so the bottom of the enclosure containing the disconnecting means is not less than ____ above finished grade or working platform.

- (a) 24" (b) 36" (c) 48" (d) 60"

53. Overhead conductors not over 1,000 volts pass over a track rails of railroads, they shall have a minimum clearance of not less than ____ feet above finished grade.

- (a) 12 (b) 15 (c) 18 (d) 24 1/2

54. Flexible cords shall be used only in continuous lengths, without splices, other than splices for the repair of hard-service cord or ____ cord under certain conditions.

- (a) junior hard-service (b) lamp (c) tinsel portable

55. Where an NM cable is installed parallel to framing members, the cable shall be installed so that the nearest edge of the cable is not less than ____ from the nearest edge of the framing member.

- (a) 1/2" (b) 1" (c) 1 1/4" (d) 1 1/2"

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56. The maximum ampere rating permitted for a 125 volt, single-phase, receptacle outlet having a cord-and-plug connected motor load that does not have individual overload protection is ____.

- (a) 15 amps (b) 20 amps (c) 25 amps (d) 30 amps

57. ____ is permitted in the dedicated electrical space above a panelboard.

- (a) Water piping (b) Leak protection (c) Sprinkler protection (d) Air-conditioning ducts

58. Which of the following is **not** required to be marked on the name plate of a transformer?

- (a) Frequency (b) Amount of insulating liquid
(c) AWG size (d) Clearance for ventilating openings

59. Branch circuits shall be rated in accordance with the ____.

- (a) ampere rating of the largest receptacle
(b) maximum permitted rating of the fuse or breaker
(c) number of receptacle outlets in the branch circuit
(d) ampere rating of the largest conductor

60. Underground wiring within ____ horizontally from the inside wall of the pool shall be permitted in liquidtight flexible metal conduit listed for direct burial use.

- (a) 18" (b) 24" (c) 48" (d) 60"

61. ____ locations are those that are hazardous because of the presence of easily ignitable fibers or flyings.

- (a) Class I (b) Class II (c) Class III (d) Class II, division II

62. You have 125 volts at the panel and 115 volts at the load. What is the percentage of voltage drop?

- (a) 5% (b) 4.35% (c) 4.17% (d) 8%

63. If the plans drawing has a scale of $1/4" = 1'$, a dimension of $3 \frac{1}{2}"$ on the drawing would be equal to ____.

- (a) 12' (b) 13' (c) 14' (d) 15'

64. When working from an electrical drawing, you should start from the ____.

- (a) lower right-hand corner (b) center (c) upper left-hand corner (d) bottom

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65. 60 cycle frequency travels 90 degrees in how many seconds?

- (a) 1/60 (b) 1/120 (c) 1/180 (d) 1/240**

66. Type MC cable shall be permitted to be unsupported and unsecured where the cable is not more than ____ in length from the last point of connection to luminaires.

- (a) 36" (b) 48" (c) 60" (d) 72"**

67. Conduit nipples not over 24" in length may be filled to a maximum of ____ of their CSA.

- (a) 50% (b) 60% (c) 70% (d) 80%**

68. What is the minimum size equipment grounding conductor for a 50 amp rated branch circuit?

- (a) #14 (b) #12 (c) #10 (d) #8**

69. Electric sign enclosures may be constructed of wood if ____.

- | | |
|-------------------------------------|---|
| (a) kept 1" from lampholders | (b) wood is not permitted at all |
| (c) kept 2" from lampholders | (d) none of these |

70. What is the full load current of a 50 horsepower, 3 ϕ , 480v, wound rotor, AC motor?

- (a) 104 amps (b) 52 amps (c) 65 amps (d) 41 amps**
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